## Type MS · Regions made by tangents and a sector

### Answer 1 of 6 · Question 4 part whole · 6 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

| Number | | |
|---|---|---|
| 4 | Area of sector $= 0.4r^2$ | B1 |
| | $BC = r\tan 0.8$ | B1 |
| | Area of triangle $= \frac{1}{2}r^2\tan 0.8$ | B1 ft |
| | Shaded region $= \frac{1}{2}r^2\tan 0.8 - 0.4r^2 = 101$<br>$r^2 = \frac{101}{\frac{1}{2}\tan 0.8 - 0.4}$<br>$r = 29.7$ | M1 M1<br>A1 |
| | | **[6]** |

| Mark | Additional Guidance |
|---|---|
| Accept angle converted to degrees $0.8^c = 45.84°$ throughout<br>$\tan(0.8) = \tan(45.8...)^0 = 1.0296$ | |
| B1 | For the correct area of the sector $= \frac{0.8}{2}r^2$ oe (need not be simplified) |
| B1 | For $BC = r\tan 0.8$ oe e.g. accept $\tan\left(\frac{4}{5}\right) = \frac{BC}{r}$<br>This may be embedded in $'\frac{r\times r\tan 0.8}{2}'-'\frac{0.8}{2}r^2' = 101$<br>Award when seen. |
| B1ft | $A = \frac{r\times r\tan 0.8}{2}$ |
| M1 | Shaded region $= '\frac{r\times r\tan 0.8}{2}'-'\frac{0.8}{2}r^2' = 101$<br>Ft their expressions for the areas of the sector and triangle provided they are as a minimum $kr^2\tan 0.8$ and $lr^2$ where $k$ and $l$ are constants |
| dM1 | **This mark is dependent on the previous M mark**<br>For attempting to solve their equation $r = \sqrt{\frac{101}{\frac{1}{2}\tan 0.8 - 0.4}} = (29.658...)$<br>**This is an A mark in Epen** |
| A1 | $r = 29.7$ only |

## Type MS · Regions made by tangents and a sector

### Answer 1 of 6 · Question 4 part whole · 6 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Regions made by tangents and a sector

Answer 2 of 6 · Question 5 part whole · 8 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part whole in this mark scheme extract

| | | |
|---|---|---|
| 5 | Triangle $OBA$: $\tan\left(\frac{\pi}{3}\right)=\frac{AB}{r}$ | M1 |
| | $AB = r\tan\left(\frac{\pi}{3}\right)$ | A1 |
| | Area of triangle $OBA = \frac{1}{2}r^2\tan\left(\frac{\pi}{3}\right)$ | M1 |
| | Area of quadrilateral $= \sqrt{3}r^2$ | A1 |
| | Area of sector $= \frac{1}{2}r^2\times\frac{2}{3}\pi=\frac{\pi}{3}r^2$ | M1 |
| | Area of shaded region $= \sqrt{3}r^2-\frac{\pi}{3}r^2=10$ oe | dM1 |
| | $r^2=\frac{10}{\sqrt{3}-\frac{\pi}{3}}$ | M1 |
| | $r = 3.82$ | A1 |
| | | **[8]** |

| | **Notes** |
|---|---|
| | Accept angles converted to degrees throughout |
| **M1** | For $\tan\left(\frac{\pi}{3}\right)=\frac{AB}{r}$ or $\tan\left(\frac{\pi}{3}\right)=\frac{AC}{r}$ |
| **A1** | For $r\tan\left(\frac{\pi}{3}\right)$ $(=AB$ or $AC)$ |
| **M1** | For finding the area of either triangle $OBA$ or $OCA = \frac{1}{2}r^2\tan\left(\frac{\pi}{3}\right)$ |
| **A1** | For finding the area of the quadrilateral $= \sqrt{3}r^2$ |
| **M1** | For finding the area of the sector $= \frac{1}{2}r^2\times\frac{2}{3}\pi$ |
| **dM1** | For area of quadrilateral – area of sector = 10 dependant on a reasonable attempt to find both the area of the quadrilateral and the area of the sector |
| **M1** | For solving the equation as far as $r^2 = ...$ |
| **A1** | For 3.82 |

## Type MS · Regions made by tangents and a sector

### Answer 2 of 6 · Question 5 part whole · 8 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part whole in this mark scheme extract

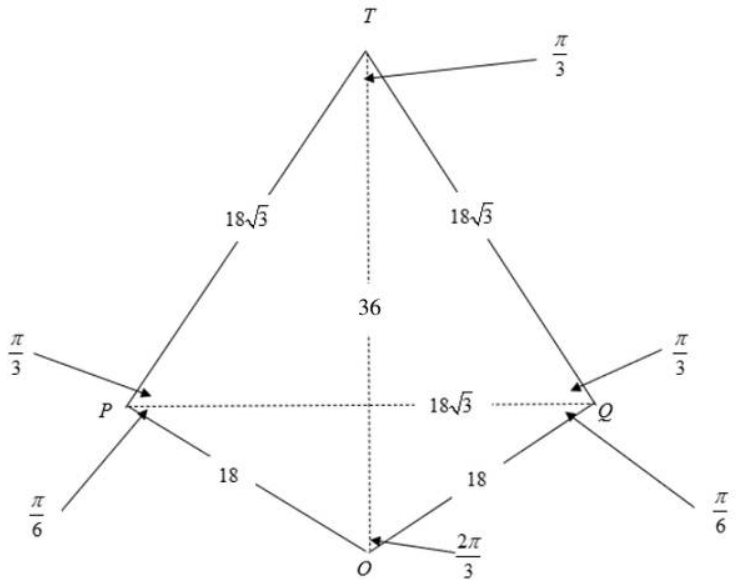
| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Regions made by tangents and a sector

### Answer 3 of 6 · Question 3 part b.i · 1 mark

2021 JUN · Paper 02 · extract 1 of 3 · find part b.i in this mark scheme extract

| | | |
|---|---|---|
| 3 (a) | $PQ = 18 \times \frac{2}{3}\pi = 12\pi$ | M1 A1 (2) |
| (b)(i) | $\alpha = \frac{1}{3}\pi$ | B1 |
| (b)(ii) | $PT = 18\tan\frac{\pi}{3} = 18\sqrt{3}$ | M1 A1 |
| | Area of $OPTQ = 2 \times \frac{1}{2} \times 18 \times 18\sqrt{3}$ | M1 |
| | Area of Sector $OPQ = \frac{1}{2} \times 18^2 \times \frac{2\pi}{3}$ | M1 |
| | Shaded Area $= 2 \times \frac{1}{2} \times 18 \times 18\sqrt{3} - \frac{1}{2} \times 18^2 \times \frac{2\pi}{3} = 222 \text{ cm}^2$ | M1 A1 (7) |
| | | **Total 9 marks** |




Area of triangle $OPQ = 81\sqrt{3}$ or 140.29... cm$^2$

Area of triangle $PQT = 243\sqrt{3}$ or 420.88... cm$^2$

Area of quadrilateral $OTPQ = 324\sqrt{3}$ or 561.18... cm$^2$

| Part | Mark | Guidance |
|---|---|---|
| **(a)** | **M1** | Uses the **correct** formula for the length of arc to give $PQ = 18 \times \frac{2}{3}\pi = ...$ |
| | **A1** | For $PQ = 12\pi$ |

Type MS · Regions made by tangents and a sector

Answer 3 of 6 · Question 3 part b.i · 1 mark

2021 JUN · Paper 02 · extract 2 of 3 · find part b.i in this mark scheme extract

| | | |
|---|---|---|
| | **ALT – works in degrees** (but the angle must be correct at $120^o$) | |
| | **M1** | Uses the **correct** formula for length of arc to give<br>$PQ = \frac{120}{360} \times 2\pi \times 18 = ...$ |
| | **A1** | For $PQ = 12\pi$ |
| **(b)** | **(i) B1** | For stating $\alpha = \frac{\pi}{3}$ (Please check the diagram as it may written on there) |
| | **(ii) Method 1** - Allow use of degrees throughout provided the angles are correct.<br>$\left(\angle POQ = 120^o,\ \angle PTQ = 60^o\right)$ | |
| | **Finds length of *PT*** | |
| | **M1** | For finding length $PT$ : e.g.,<br>$\tan\left(\frac{\pi}{3}\right) = \frac{PT}{18} \Rightarrow PT = 18\tan\left(\frac{\pi}{3}\right) = ...$<br>The given values must be used correctly |
| | **A1** | For $PT = 18\sqrt{3}$ |
| | **M1** | For the area of $OPTQ = 2 \times \frac{1}{2} \times 18 \times '18\sqrt{3}' = \left(561.18...\right)$<br>Their $18\sqrt{3}$ must come from an attempt at using trigonometry. |
| | **Method 2** | |
| | **Finds lengths *PQ* and *TO*** | |
| | **M1** | For finding the lengths $PQ$ **and** $TO$ using any acceptable correct trigonometry.<br>e.g., $PQ = \sqrt{18^2 + 18^2 - 2 \times 18 \times 18\cos\left(\frac{2\pi}{3}\right)} = ...$ and<br>$TO = \frac{\cos\left(\frac{\pi}{3}\right)}{18} = ...$<br>The given values must be used correctly |
| | **A1** | For both correct lengths: $PQ = 18\sqrt{3}$ and $TO = 36$ |
| | **M1** | For the area of $OPTQ = \frac{1}{2} \times '18\sqrt{3}' \times '36' = \left(561.18...\text{ or } 324\sqrt{3}\right)$<br>Their $18\sqrt{3}$ and $36$ must come from an attempt at using trigonometry |
| | | |
| | **M1** | For the area of sector $OPQ = \frac{1}{2} \times 18^2 \times \frac{2\pi}{3} = \left(339.29...\text{ or } 108\pi\right)$<br>or $\frac{120^o}{360^o} \times \pi \times 18^2 = \left(339.29...\right)$ |
| | **M1** | For area of $OPTQ$ – area of Sector $OPQ$<br>$561.18 - 339.29 = 221.887\ldots$ |
| | **A1** | For $222\ \text{cm}^2$ (must be 3sf) (Units are not required) |

## Type MS · Regions made by tangents and a sector

### Answer 3 of 6 · Question 3 part b.i · 1 mark

2021 JUN · Paper 02 · extract 3 of 3 · find part b.i in this mark scheme extract

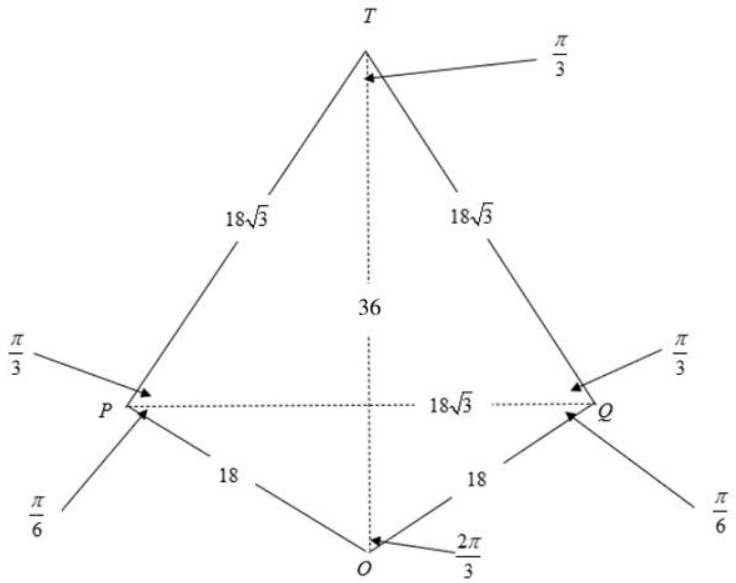
| Question number | Scheme | Marks |
|---|---|---|

## Type MS · Regions made by tangents and a sector

### Answer 4 of 6 · Question 3 part b.ii · 6 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part b.ii in this mark scheme extract

| | | |
|---|---|---|
| 3 (a) | $PQ = 18 \times \frac{2}{3}\pi = 12\pi$ | M1 A1 (2) |
| (b)(i) | $\alpha = \frac{1}{3}\pi$ | B1 |
| (b)(ii) | $PT = 18\tan\frac{\pi}{3} = 18\sqrt{3}$ | M1 A1 |
| | Area of $OPTQ = 2 \times \frac{1}{2} \times 18 \times 18\sqrt{3}$ | M1 |
| | Area of Sector $OPQ = \frac{1}{2} \times 18^2 \times \frac{2\pi}{3}$ | M1 |
| | Shaded Area $= 2 \times \frac{1}{2} \times 18 \times 18\sqrt{3} - \frac{1}{2} \times 18^2 \times \frac{2\pi}{3} = 222 \text{ cm}^2$ | M1 A1 (7) |
| | | **Total 9 marks** |




Area of triangle $OPQ = 81\sqrt{3}$ or 140.29... cm$^2$

Area of triangle $PQT = 243\sqrt{3}$ or 420.88... cm$^2$

Area of quadrilateral $OTPQ = 324\sqrt{3}$ or 561.18... cm$^2$

| Part | Mark | Guidance |
|---|---|---|
| **(a)** | **M1** | Uses the **correct** formula for the length of arc to give $PQ = 18 \times \frac{2}{3}\pi = ...$ |
| | **A1** | For $PQ = 12\pi$ |

Type MS · Regions made by tangents and a sector

Answer 4 of 6 · Question 3 part b.ii · 6 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part b.ii in this mark scheme extract

| | | |
|---|---|---|
| | **ALT – works in degrees** (but the angle must be correct at $120^o$) | |
| | **M1** | Uses the **correct** formula for length of arc to give $PQ = \frac{120}{360} \times 2\pi \times 18 = ...$ |
| | **A1** | For $PQ = 12\pi$ |
| **(b)** | **(i) B1** | For stating $\alpha = \frac{\pi}{3}$ (Please check the diagram as it may written on there) |
| | **(ii) Method 1** - Allow use of degrees throughout provided the angles are correct. $\left(\angle POQ = 120^o,\ \angle PTQ = 60^o\right)$ | |
| | **Finds length of *PT*** | |
| | **M1** | For finding length $PT$ : e.g., $\tan\left(\frac{\pi}{3}\right) = \frac{PT}{18} \Rightarrow PT = 18\tan\left(\frac{\pi}{3}\right) = ...$ The given values must be used correctly |
| | **A1** | For $PT = 18\sqrt{3}$ |
| | **M1** | For the area of $OPTQ = 2 \times \frac{1}{2} \times 18 \times '18\sqrt{3}' = \left(561.18...\right)$ Their $18\sqrt{3}$ must come from an attempt at using trigonometry. |
| | **Method 2** | |
| | **Finds lengths *PQ* and *TO*** | |
| | **M1** | For finding the lengths $PQ$ **and** $TO$ using any acceptable correct trigonometry. e.g., $PQ = \sqrt{18^2 + 18^2 - 2 \times 18 \times 18\cos\left(\frac{2\pi}{3}\right)} = ...$ and $TO = \frac{\cos\left(\frac{\pi}{3}\right)}{18} = ...$ The given values must be used correctly |
| | **A1** | For both correct lengths: $PQ = 18\sqrt{3}$ and $TO = 36$ |
| | **M1** | For the area of $OPTQ = \frac{1}{2} \times '18\sqrt{3}' \times '36' = \left(561.18...\text{ or } 324\sqrt{3}\right)$ Their $18\sqrt{3}$ and $36$ must come from an attempt at using trigonometry |
| | | |
| | **M1** | For the area of sector $OPQ = \frac{1}{2} \times 18^2 \times \frac{2\pi}{3} = \left(339.29...\text{ or } 108\pi\right)$ or $\frac{120^o}{360^o} \times \pi \times 18^2 = \left(339.29...\right)$ |
| | **M1** | For area of $OPTQ$ – area of Sector $OPQ$ $561.18 - 339.29 = 221.887\ldots$ |
| | **A1** | For $222\ \text{cm}^2$ (must be 3sf) (Units are not required) |

## Type MS · Regions made by tangents and a sector

### Answer 4 of 6 · Question 3 part b.ii · 6 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part b.ii in this mark scheme extract

| Question number | Scheme | Marks |
|---|---|---|

Type MS · Regions made by tangents and a sector

Answer 5 of 6 · Question 3 part a · 2 marks

2024 JUN · Paper 02 · extract 1 of 3 · find part a in this mark scheme extract

| | |
|---|---|
| B1 | $\left[\left(\frac{-}{3}\right)^{-}2\times\frac{-}{3}\right]$ seen anywhere in their working.<br>This can be embedded anywhere in their work<br>This can also be implied from a correct value of the SUM $\frac{\alpha}{2\beta}+\frac{\beta}{2\alpha}=\frac{19}{6}$ |
| B1 | For the correct PRODUCT of $\frac{1}{4}$ |
| M1 | For the correct algebra $\left[\frac{2\alpha^2+2\beta^2}{4\alpha\beta}\right]$ **and** substitution of their values of $\alpha^2+\beta^2$ [even if the algebra on $\alpha^2+\beta^2$ is incorrect] to find the SUM.<br>Ft their $\frac{19}{9}$ even if it comes from incorrect algebra on $\alpha^2+\beta^2$ earlier. |
| A1 | For the value of $\frac{19}{6}$ |
| M1 | For a correct equation in any form.<br>Need not = 0 for this mark |
| A1 | For a correct equation with integer coefficients = 0 |

Type MS · Regions made by tangents and a sector

Answer 5 of 6 · Question 3 part a · 2 marks

2024 JUN · Paper 02 · extract 2 of 3 · find part a in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|
| 3(a) | $\sin\left(\frac{\theta}{2}\right)=\frac{r}{2r}\Rightarrow\frac{\theta}{2}=\frac{\pi}{6}$ | M1 |
| | $\Rightarrow\theta=\frac{\pi}{3}$ | A1<br>[2] |
| | **ALT works in degrees** | |
| | $\sin\left(\frac{\theta}{2}\right)=\frac{r}{2r}\Rightarrow 30^{\circ}$ | [M1 |
| | $\Rightarrow\theta=\frac{\pi}{3}$ | A1] |
| (b) | Area of circle = $\pi r^2$ | B1 |
| | Area of sector = $\frac{\pi}{3}\times(3r)^2=\frac{3}{2}\pi r^2$ | M1 |
| | Shaded area: $8\pi=\frac{3}{2}\pi r^2-\pi r^2=\frac{\pi r^2}{2}\Rightarrow r^2=16\Rightarrow r=4$ (cm) | dM1A1<br>[4] |
| | **ALT works in degrees** | |
| | Area of circle = $\pi r^2$ | [B1 |
| | Area of sector = $\frac{60^{\circ}}{360^{\circ}}\times\pi\times(3r)^2=\frac{3}{2}\pi r^2$ | M1 |
| | Shaded area: $8\pi=\frac{3}{2}\pi r^2-\pi r^2=\frac{\pi r^2}{2}\Rightarrow r^2=16\Rightarrow r=4$ (cm) | dM1A1] |
| | | **Total 6 marks** |

## Type MS · Regions made by tangents and a sector

### Answer 5 of 6 · Question 3 part a · 2 marks

2024 JUN · Paper 02 · extract 3 of 3 · find part a in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|

Type MS · Regions made by tangents and a sector

Answer 6 of 6 · Question 3 part b · 4 marks

2024 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

| | |
|---|---|
| B1 | $\left[\left(\frac{-}{3}\right)^{-2\times\frac{-}{3}}\right]$ seen anywhere in their working.<br>This can be embedded anywhere in their work<br>This can also be implied from a correct value of the SUM $\frac{\alpha}{2\beta}+\frac{\beta}{2\alpha}=\frac{19}{6}$ |
| B1 | For the correct PRODUCT of $\frac{1}{4}$ |
| M1 | For the correct algebra $\left[\frac{2\alpha^2+2\beta^2}{4\alpha\beta}\right]$ **and** substitution of their values of $\alpha^2+\beta^2$ [even if the algebra on $\alpha^2+\beta^2$ is incorrect] to find the SUM.<br>Ft their $\frac{19}{9}$ even if it comes from incorrect algebra on $\alpha^2+\beta^2$ earlier. |
| A1 | For the value of $\frac{19}{6}$ |
| M1 | For a correct equation in any form.<br>Need not = 0 for this mark |
| A1 | For a correct equation with integer coefficients = 0 |

Type MS · Regions made by tangents and a sector

Answer 6 of 6 · Question 3 part b · 4 marks

2024 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|
| 3(a) | $\sin\left(\frac{\theta}{2}\right)=\frac{r}{2r}\Rightarrow\frac{\theta}{2}=\frac{\pi}{6}$ | M1 |
| | $\Rightarrow\theta=\frac{\pi}{3}$ | A1<br>[2] |
| | **ALT works in degrees** | |
| | $\sin\left(\frac{\theta}{2}\right)=\frac{r}{2r}\Rightarrow 30^{\circ}$ | [M1 |
| | $\Rightarrow\theta=\frac{\pi}{3}$ | A1] |
| (b) | Area of circle = $\pi r^2$ | B1 |
| | Area of sector = $\frac{\pi}{3}\times(3r)^2=\frac{3}{2}\pi r^2$ | M1 |
| | Shaded area: $8\pi=\frac{3}{2}\pi r^2-\pi r^2=\frac{\pi r^2}{2}\Rightarrow r^2=16\Rightarrow r=4$ (cm) | dM1A1<br>[4] |
| | **ALT works in degrees** | |
| | Area of circle = $\pi r^2$ | [B1 |
| | Area of sector = $\frac{60^{\circ}}{360^{\circ}}\times\pi\times(3r)^2=\frac{3}{2}\pi r^2$ | M1 |
| | Shaded area: $8\pi=\frac{3}{2}\pi r^2-\pi r^2=\frac{\pi r^2}{2}\Rightarrow r^2=16\Rightarrow r=4$ (cm) | dM1A1] |
| | | **Total 6 marks** |

## Type MS · Regions made by tangents and a sector

### Answer 6 of 6 · Question 3 part b · 4 marks

2024 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

| Question | Scheme | Marks |
|---|---|---|